SUPPORTING INFORMATION

Calculation of multiphase equilibria containing mixed solvents and mixed electrolytes: General formulation and Case studies.

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1) ePC-SAFT framework

The EoS ePC-SAFT advanced[S1,S2] was used in this work to model the phase behavior of the investigated systems. Like the previous electrolyte version ePC-SAFT revised[S3] and the original PC-SAFT[S4], the thermodynamic behavior of the system is expressed as the residual Helmholtz energy at given values of temperature T, molar volume v (or density rho) and composition $\bar{x}$ (Eq. (S1)).

$a^{res} = a^{res}(T, v, \bar{x})$	(S1)

The residual Helmholtz energy is formulated as a sum of different contributions. In addition to the contribution due to original PC-SAFT, which are the hard-chain a^{HC}, the dispersion a^{disp} and the association a^{assoc} contribution, ePC-SAFT advanced includes the Debye-Hückel a^{DH} and the Born term a^{Born} in the final expression of the residual Helmholtz energy (Eq. (S2)).

$a^{res} = a^{HC} + a^{disp} + a^{assoc} + a^{DH} + a^{Born}$	(S2)

In Eq. (S2), each term has a given physical meaning and aims at representing a specific molecular interaction. The hard-chain term a^{HC} accounts for the repulsions forces due to the own volume of the molecules and for their non-spherical form. The dispersion term a^{disp} accounts for short-range anisotrope attractive forces among the molecules, such as Van der Waals or weak dipolar forces. The association term a^{assoc} accounts for short-range strong and highly directional forces such as hydrogen bonds. The Debye-Hückel a^{DH} and the Born term a^{Born} are needed to describe intermolecular interactions due to charged species in the system: the Debye-Hückel term a^{DH} accounts for the long-range electrostatic forces among ions whereas the Born term a^{Born} describes electrostatic ion-dipole interactions of the ions with the surrounding (charged and non-charged) components. The value of the fugacity coefficient φ_i of each component $i = 1, \ldots, N$ can be obtained from the residual Helmholtz energy provided by ePC-SAFT advanced using Eq. (S3).

$$\ln(\varphi_i) = \frac{\mu_i^{res}}{k_B \cdot T} - \ln\left(1 + \left(\frac{\partial\left(\frac{a^{res}}{k_B \cdot T}\right)}{\partial \rho}\right)\right)$$	(S3)

Like ePC-SAFT revised and the original PC-SAFT, ePC-SAFT advanced requires five pure-component parameters for each component: the segment diameter σ_i, the number of segments m_i^{seg}, the dispersion-energy parameter u_i, the association-energy parameter $\varepsilon^{A_iB_i}$ and association-volume parameter $\kappa^{A_iB_i}$. Furthermore, for each pair of components *i* and *j* a binary interaction parameter k_{ij} is required. Formally, this corrects for deviations from the adopted combining rules for the dispersion energy between different components and is used to obtain quantitative agreement with experimental data. The used combining rules of Berthelot Lorentz are given by Eqs. (S4) – (S6).

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j)(1 - l_{ij})$$	(S4)
$$u_{ij} = \sqrt{u_i u_j}(1 - k_{ij}(T))$$	(S5)
$$\varepsilon^{A_iB_j} = \frac{\varepsilon^{A_iB_i} + \varepsilon^{A_jB_j}}{2}(1 - k_{ij}^{hb})$$	(S6)

The binary interaction parameter $k_{ij}(T)$ is described as a linear function of the temperature according to Eq (S7).

$$k_{ij}(T) = k_{ij,298.15} + k_{ij\_T}(T - 298.15)$$	(S7)

The binary interaction parameters l_{ij} and k_{ij}^{hb} from Eqs. (S4) and (S6) are usually set to zero. In this work, they were used only between water and 1-butanol to improve the agreement between calculated and experimental binary LLE.

2) ePC-SAFT parameters

The pure-component parameters used in this work to model non-charged components are listed in Table S1.

Table S1: Pure-component parameters of the non-charged components used in this work.

component	m_i^{seg}	σ_i / Å	u_i/k_B / K	ε^{AiBi} / K	κ^{AiBi}	Ref.
Water	1.2047	*	353.95	2425.7	0.04509	S5
n-hexane	3.0578	3.7983	236.77	-	-	S4
n-dodecane	5.3060	3.8959	249.21	-	-	S4
1-butanol	2.7510	3.6139	259.59	2544.56	0.00669	S6
1-propanol	3.0000	3.2522	233.40	2276.78	0.01527	S7
$* \sigma = 2.7927 + \left(10.11 \cdot e^{-0.01775\,T} - 1.417 \cdot e^{-0.01146\,T}\right)$						

Pure-component parameters used to model ions are listed in Table S2.

Table S2: Pure-component parameters of the charged species used in this work.

Ionic species	m_i^{seg}	σ_i / Å	u_i/k_B / K	Ref.

$[P_{66614}]^+$	1.4872	3.5926	206.49	S8
$[DCA]^-$	3.7432	3.8771	509.31	S9
$[NH_4]^+$	1	3.5740	230.00	S10
$[K]^+$	1	3.3417	200.00	S10
$[Na]^+$	1	2.8232	230.00	S10
$[Cl]^-$	1	2.7560	170.00	S10

The binary interaction parameters between water and the organic solvents used in this work are listed in Table S3.

Table S3: Binary interaction parameters between water and the organic solvents used in this work.

	1-butanol	1-pentanol
$k_{ij\_298.15K}$	-0.102	-0.017
k_{ij_T}	$2.94\text{x}10^{-4}$	0
l_{ij}	-0.0044	0
k_{ij}^{hb}	0.026	0
Ref.	S6	S11

Binary interaction parameters between ions were taken from Held *et al.*[S3] and are shown in Table S4.

Table S4: Binary interaction parameters $k_{ij\_298.15K}$ between the cations and the chloride anion used in this work.

$k_{ij\_298.15K}$	Na^+	K^+	NH_4^+
Cl^-	0.317	0.064	-0.566

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